

Bond Futures Contracts

FUTURES PRICE AND DELIVERY OPTION

The key difference between a bond forward and a bond futures contract is the existence of a *set* of deliverable bonds, called the *deliverable basket*. In particular, the person with a short position in a bond futures contract (i.e. the short) has a choice of bonds he can deliver. Since the bonds in the deliverable basket have different coupons and maturity dates, the exchanges introduce the concept of a *conversion factor* (CF), which is applied to the price of each bond when determining the invoice price that the short receives when delivering a particular bond into a particular futures contract. In order to make all deliverable bonds comparable, a CF is defined for each, which roughly corresponds to one-hundredth of the price of the deliverable bond at a certain yield level (often 6%). This yield level is called the *notional coupon* of the bond futures contract.

Technical Points Regarding the Conversion Factor

- The CF is a constant for each individual deliverable bond.
- The CF of a bond will depend on the expiration date of the futures contract. For example, if a bond is deliverable into both the March and the June contracts, it will have a different CF for each expiration month.
- In general terms, the CF of a bond is one-hundredth of the price the bond would have at expiration of the futures contract if the bond yield was equal to the notional coupon for that contract.
- The notional coupon of the futures contract is defined at the time the contract is first listed by the exchange. Like the CF, once the notional coupon for a particular expiration month is defined, it does not change.

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- However, exchanges can and do sometimes switch to a new notional coupon when listing a new futures contract, particularly when the yields of the bonds in the deliverable basket are far from the notional coupon used most recently.
- Changes to the notional coupon can have a material effect on the value of the delivery option and the value of the futures contract, as discussed below.
- Note that the calculation of the CF as specified in the contract documentation may not correspond exactly to one-hundredth of the bond price at the yield of the notional coupon. The exchanges publish formulae that analysts can use to predict the CF before a contract is listed, but the analyst should be careful to confirm the CF for each contract when it is listed, as the CF published at the time of listing is determinative, irrespective of any formulae the exchange may have published to assist analysts.

The Delivery Process

At delivery, the person with the short position in the bond futures contract can choose to deliver any bond in the deliverable basket and will receive a payment of the futures price, F , multiplied by the CF for the bond he has chosen, in exchange for delivery of the bond. Thus, if the yield at delivery equals the notional coupon of the bond futures contract, and if the CFs represent exactly one-hundredth of the prices of the bonds at that yield level, then the futures contract should trade at 100, and the prices of all deliverable bonds should be equal to the compensation received for their delivery into the futures contract. In that case, the short would be indifferent about the bond he chooses to deliver. In general, however, there are differences with regards to the payoffs at delivery between the deliverable bonds. Assuming the short buys a deliverable bond right at delivery at price P , and delivers it immediately into the futures contract, his P&L will be $F \times CF - P$. If there are n deliverable bonds in the basket with prices $P_1, \dots, P_n$ and conversion factors $CF_1, \dots, CF_n$, he can compare the payoffs for buying and delivering each of them with a table like this:

$$\text{Bond 1: } F \times CF_1 - P_1$$

...

$$\text{Bond } n: F \times CF_n - P_n$$

The deliverable bond with the best payoff (i.e. the largest number in the table above) is called the *cheapest-to-deliver* (CTD), and the short will usually

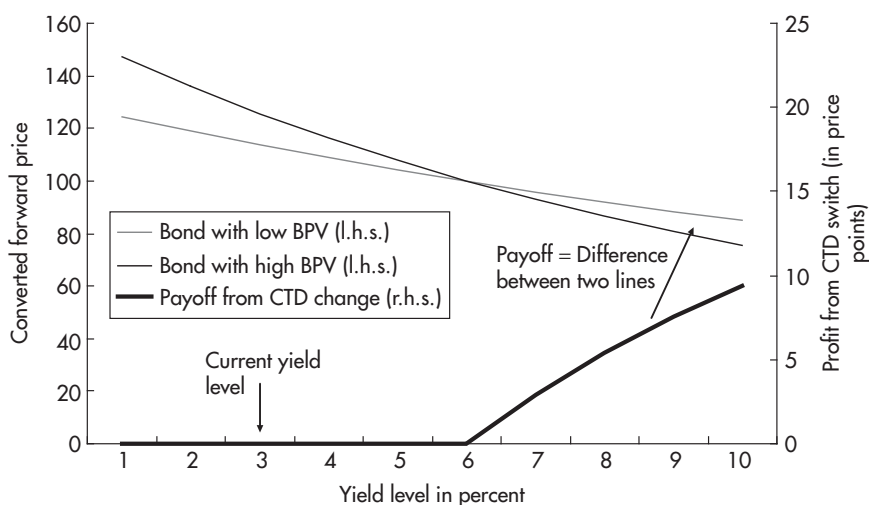


FIGURE 7.1 CTD situation as a function of the yield level.

Source: Authors.

deliver that bond. If the yields of all deliverable bonds at delivery were equal to the notional coupon of the bond futures contract, and if the CF of each bond was exactly equal to one-hundredth of the price of the bond at that yield level, then all deliverable bonds would be equally CTD.¹ At higher yield levels, only the bonds with largest basis point values (BPVs) in the basket tend to be CTD, while at lower yield levels only the bonds with the smallest BPV in the basket tend to be CTD. The reason for this behavior is illustrated in Figure 7.1. When the general level of yields is high, the converted prices (P/CF) of large BPV bonds tend to be low relative to the converted prices of bonds with a smaller BPV. If the two bonds of Figure 7.1 were the only ones that were deliverable, the one with the lower converted price would be CTD at a given yield level. Thus, the CTD would switch between the bond with a small BPV to the bond with a large BPV at the notional coupon (6% in this example). This switching point is also called the *inflection point*.

CTD Switches and the Delivery Option

Since the short has the right to choose the bond he delivers, he can profit from changes in the CTD. The monetary value of that right is called the *delivery*

¹In this scenario, a difference in coupons would result in some bonds being rich relative to others, given the coupon effect, perhaps even allowing reconstitution arbitrage (via the strip market).

option (DO) value of the bond futures contract. Using the illustration from above as an example, imagine we are three months before delivery and at a yield level of 3%. At that yield level, the bond with a small BPV is CTD, so we could buy that bond on a forward basis and sell the futures contract. If at delivery the same bond is still CTD, we can deliver it and close our positions at zero cost. If, however, a rise in yields has caused the CTD to switch to the bond with a large BPV, we could sell our bond, buy the new CTD, and deliver this cheaper bond into our short position in the futures contract. A switch in CTD is equivalent to a profit from our bond switch operation, with the amount of profit given by the difference of the converted forward prices of the two bonds (the difference of the two lines in Figure 7.1).

To prevent arbitrage at delivery, the futures contract must settle at a price such that there is no profit or loss from buying the CTD in the bond market and delivering it into the contract. Thus, at delivery, the equation $F \times CF_{CTD} - P_{CTD} = 0$ holds and implies a fair futures price of $\frac{P_{CTD}}{CF_{CTD}}$. Likewise, before delivery, the fair futures price is a function of the converted forward price of the CTD. But as the CTD can still change, it needs to be adjusted for the DO value, and the fair futures price is given as $\frac{FwdP_{CTD}}{CF_{CTD}} - DO$.

Net and Gross Bond Basis

That arbitrage relationship at delivery is the key for basis trades between deliverable bonds and the future, with the following conventions:

$$\text{Gross basis} = P - F \times CF$$

$$\text{Net basis} = FwdP - F \times CF$$

Note that the fair value of the CTD net basis is equal to $DO \times CF_{CTD}$. Thus, if there is no DO, in particular at the delivery date, then the fair value of the CTD net basis is 0. And as the DO is never negative, the fair value of any net basis is always greater than or equal to zero.²

From these equations, it is clear that the DO is key to pricing and trading bond futures contracts. All other aspects of the contract can be hedged in the markets. For example, the forward price of a deliverable bond is a direct function of the bond and of the repo market. (The CF is a pre-defined constant.) Thus, the model used to calculate the fair value of the DO will determine the

²At times, specific situations like penalties for failing to deliver might result in a negative net basis being observed in the market.

fair value of the futures contract and therefore whether basis trades are attractive. In addition, the better the DO is understood, the better the futures contract can be applied for hedging purposes, and the more informed the appropriate time for rolling over front into back month contracts can be decided.

Imagine that the net basis of the CTD quotes in the market at 5 cents and that the CF of the CTD is 1. If the model used to evaluate the DO returns a fair DO price of 2 cents, selling the CTD net basis (i.e. selling the CTD in the forward market and buying the futures contract) appears attractive. If another model returns a fair DO price of 10 cents, however, buying the CTD net basis looks profitable. This illustrates the crucial importance of the model used to price the DO for correctly evaluating the fair value of the futures contract and for ensuring that one is on the right side of basis trades.

As the futures price observed in the market in some sense should reflect the average pricing model used by market participants, having a better model enables us to exploit the shortcoming of the average model. The CTD net basis in the market is a function of the DO models of the other market participants, which we can compare with the results of our DO model. If we have a superior DO model and the market price is not in line with its results, we can translate the superiority of our model directly into profitable trading strategies by entering into CTD net basis positions. For example, if the DO model used by market participants gives a fair value of the DO of 5 cents, and if the CF of the CTD is 1, then the CTD net basis should trade at 5 cents. If our better model gives us a fair value for the DO of 10 cents, then we can translate the theoretical advantage of our model into trading profit by buying the CTD net basis in a market that underestimates its value.

We shall therefore describe two different DO models. The first one is the one-factor DO model commonly used in the market, while the second one is a superior multi-factor DO model we developed while working at ABN Amro. The model arbitrage mentioned in the previous paragraph will therefore consist in assessing the market price for the DO (CTD net basis/CF), calculated by one-factor models, through the lens of a multi-factor model.

ONE-FACTOR DELIVERY OPTION MODELS³

Figure 7.1 can serve as the basis for a straightforward DO model. Since the payoff profile from a CTD switch is a function of the yield level, the value of the DO can be estimated by using an option pricing model.

³To our knowledge, that model was first described in *The Treasury Bond Basis* by Galen Burghardt, Terry Belton, Morton Lane, and John Pappa, published by McGraw-Hill in 1989.

- The payoff profile of the DO is almost that of a put or call.⁴ In the example of Figure 7.1, buying the CTD net basis is like buying a fraction of puts on the future. Note that this relationship in principle allows trading the CTD net basis versus options on the bond futures contract. Usually, however, the CTD switch situation is more complex than in Figure 7.1, which makes that arbitrage rarely practicable, as discussed below.
- The strike price of the DO is determined by the CTD inflection point. Note that the DO is always an OTM (out-of-the-money) option, as the CTD needs to change (i.e. something needs to happen) for the DO to produce profit. This means that the DO has only time value and is the reason that the DO value is equal to zero at the delivery date.
- The market consensus about yield level volatility until delivery is reflected in the option prices quoted in the market (e.g. from the options on bond futures).

With this information, the DO can be calculated through an option model like Black–Scholes, using the payoff profile from Figure 7.1 and the expected volatility of the market until delivery (e.g. as reflected in the bond futures option quotes) as inputs. Qualitatively, the DO value will be large, if:

- The CTD switching point is close to the current yield level. This corresponds to the strike price of an OTM option being close to the current market price. In this case, not a lot needs to happen for the CTD to change and there is a high likelihood of a CTD switch occurring before delivery.
- The delivery date is far away. In this case, there is considerable time for a CTD switch to occur and thus there is a high likelihood of a CTD switch occurring before delivery.
- The volatility is high. In this case, there is a high likelihood of a CTD switch occurring before delivery.
- The difference between the converted BPV of the old CTD and that of the new CTD (after a switch takes place) is large. In this case, a given amount of change in yield level causes a high profit from switching out of the old into the new CTD. In Figure 7.1, a large difference between the converted BPVs corresponds to a large difference in the slope of the converted forward price lines and hence to a larger payoff in the case of a CTD switch. Therefore, the profit from a CTD switch tends to be an increasing function of the difference between the converted forward BPVs.

⁴Due to convexity effects, the relationship of the payoff to the yield level is not exactly linear.

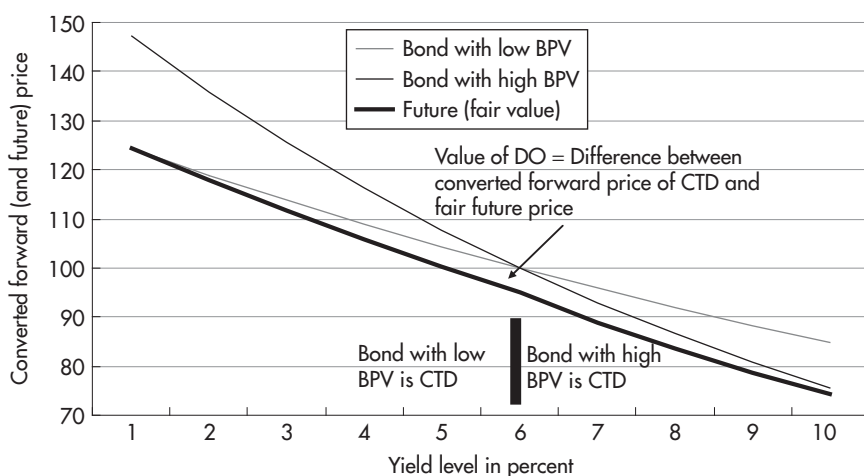


FIGURE 7.2 Fair futures price as a function of the yield level.

Source: Authors.

The first feature of our list means that for a given bond futures contract at a given date, the DO value is largest when the price of the futures contract is at the CTD inflection point. In other words, the impact on the fair futures price is a decreasing function of the distance between the yield of the current CTD and the inflection point. This result is illustrated in Figure 7.2.

Given the use of futures contracts for hedging bonds, an important application of this model is the calculation of the correct BPV for the futures contract. Simply assuming that the BPV of the futures contract is equal to the converted forward BPV of the CTD, $\frac{FwdBPV_{CTD}}{CF_{CTD}}$, ignores the impact of the DO on the futures price and might only be an acceptable approximation, particularly when the yield of the CTD is far from the inflection point. In addition, that number changes in a discontinuous fashion at the inflection point, which is a major problem for bond hedges using futures contracts. It is therefore necessary to use the slope of the 'Future (fair value)' line in Figure 7.2 as the BPV for the futures contract in hedging operations. That number is called the option-adjusted BPV (OABPV) and can be obtained by calculating the fair value of the future (i.e. $\frac{FwdP_{CTD}}{CF_{CTD}} - DO$) at two slightly different yield levels, thereby taking the impact of the change of the yield level on the DO into account.

Since changes in the absolute yield level may impact different deliverable bonds to a different extent, one should adjust the calculation for this effect, for example, by using the betas of a regression of the yield level of a deliverable bond versus the yield level of the CTD. The DO calculation then looks like the one shown in Box 7.1.

Box 7.1**One-Factor Delivery Option Model**

- For a reasonable range of yield levels for the CTD at delivery, calculate the yield levels of all deliverable issues at delivery (e.g. by using the betas from a regression).
- Determine the inflection points for CTD switches and the payoff from CTD switches.
- Decompose that payoff into a combination of puts and calls, as illustrated in Figure 7.1.
- For each of these puts and calls, obtain the relevant implied volatility from the futures options market. In order to account for the skew, use a futures option with a strike close to the yield level, at which the CTD switch occurs (i.e. the 'strike' of the put or call from Figure 7.1). Thus, each of the puts and calls are priced with an individual implied volatility.
- For each of these puts and calls, calculate the value. This could be done by using the implied volatility and 'strike' levels as input into the Black-Scholes formula.
- The DO is the sum of the option prices obtained in the previous step.
- Alternatively, the payoff function (Figure 7.1) could be numerically integrated over the probability density for the CTD yield, which again can be obtained from the futures options market. This alternative is preferable in case of the payoff function exhibiting a significant convexity. On the other hand, as it is easier to account for the skew through individual puts and calls rather than through an adjustment of the probability density for the CTD yield, this alternative is not advisable in the presence of a large skew.

THE NEED FOR MULTI-FACTOR DELIVERY OPTION MODELS

By construction, the DO model described above is a one-factor model. It can only assess the impact of changes in the absolute yield level on the DO; it cannot consider CTD switches that occur due to reasons other than a change in the overall yield level. Thus, those CTD switches that happen without a change in absolute yield level are outside of the scope of that model, which may therefore underestimate the value of the DO. In other words, as the model only

assesses the DO coming from changes in the absolute yield level, it is conceptually ignorant of DO value coming from other sources, for example, from a bond becoming CTD due to only its individual repo rate decreasing.

In general, the DO value arises from the volatilities of the differences between the prices of deliverable bonds, which thereby determine the likelihood of and profit from CTD switches. The prices of the deliverable bonds, and hence the differences in prices, can be modeled indirectly by modeling the yields of the deliverable bonds.

In the one-factor model discussed above, all changes in the yield spreads between deliverable bonds were the result of the different yield betas associated with each bond. When our main yield factor increased, the yields of bonds with large betas would increase by more than would the yields of bonds with smaller betas. So while the one-factor model is capable of producing varying yield spreads, the entirety of the variation in yield spreads is the result of an increase or decrease in the general level of yields.

But while the results of our PCA chapter indicate that roughly 95% of the variation in yields generally can be attributed to a single factor (see Figure 3.4), it is *not* the case that the vast majority of the variation in yield *spreads* can be attributed to a single factor. In fact, the proportion of yield *spread* variation that can be attributed to changes in the overall level of yields depends on the situation.

- If the market yield level is close to the notional coupon of the bond futures contract, CTD switches between the short end and the long end of the basket can occur. These switches are usually caused by a change in the overall yield level and therefore captured by a one-factor model. Correspondingly, they show up in Figure 7.1 and Figure 7.2, which depict the perspective of one-factor models.

As explained above, these CTD switches between the short and long end of the basket result in a large DO value due to the big difference in converted BPVs. Thus, when the market yield level is close to the notional coupon of the bond futures contract, one-factor models can be expected to capture the most important source of the DO value.

- The further the market yield level is away from the notional coupon of the bond futures contract, the less likely those CTD switches between the long and short end of the basket become. As a consequence, the DO value calculated by a one-factor model decreases and converges to zero. In the example of Figure 7.1, at a yield level of 3%, it is extremely unlikely to reach the 6% needed for a CTD switch caused by changes in the overall yield level in the few months until delivery, and thus the model based on Figure 7.1 will return a negligible DO value.